TASK SPECIFICATION: P5-07 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

Parking Booking & Encrypted Pass Modal UI

Git Branch: `feature/P5-07-booking-ui` · Target Pull Request: feature/P5-07-booking-ui

Core Objective & Technical Scope:
Reservation modal with duration sliders, vehicle picker, live tariff calculation, and scannable QR gate pass display.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-07-booking-ui`
Upstream Dependencies	P5-03, P5-06		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Build BookingModal with duration slider, vehicle selection, and live cost calculation.
2. Wire reservation submission to Booking API with optimistic UI updates.
3. Build DigitalPassModal rendering encrypted QR code and booking reference.
4. Handle error cases (insufficient funds, slot conflict).

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/features/parking/booking/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Calculated price matches server-side tariff calculation exactly.
- QR pass renders crisply and scans cleanly with standard barcode scanners.

VERIFICATION & TEST COVERAGE STRATEGY

- BookingModal validation test
- PassModal QR test
- Playwright booking flow